

Reinforcement Learning Based Traffic Signal Control: A Performance Comparison Under Different Traffic Scenarios

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Abstract Traffic congestion, which is increasing in megacities in parallel with population growth and urbanization, has become one of the most important problems in modern urban transportation. In this context, the effectiveness of control strategies at signalized intersections is even more important. In this study, a comparative evaluation of different traffic signal control strategies was conducted in a single intersection scenario using the SUMO Urban Mobility Model. The main objective of the study is to demonstrate the effectiveness of the reinforcement learning (RL) approach in adaptive traffic signal control, especially in comparison with traditional and rule-based methods. In this context, four different control methods were implemented and tested: fixed-time control, vehicle-initiated (actuated) control specific to the SUMO platform, a fuzzy logic controller and an RL-based controller. The RL model was trained on episodes of different durations (50, 150, 300 and 500) to study the performance changes during the training process. The simulations were conducted under two different demand scenarios: low traffic density (3196 vehicles) and high traffic density (6748 vehicles). The average waiting time and average travel time were used to evaluate system performance. It can be observed that the RL-based method has poor performance when traffic volume is low but holds an overwhelming advantage in comparison with other control methods when traffic volume is high.

Keywords: traffic signal control, reinforcement learning, SUMO,

I. INTRODUCTION

Global population growth and the rapid upward trend of urbanization have made traffic congestion in megacities one of the matters to be addressed urgently [1]. It results in operating inefficiencies such as increased travel time and fuel usage but significantly impacts environmental pollution from vehicle-based CO₂ emissions [2]. Traditional ways of expanding transport infrastructure, such as building new roads or laying new lanes on current roads, generally fail to provide realistic and sustainable alternatives due to exorbitant costs and physical limitations in cities [1], [3]. Therefore, intelligent transport systems (ITS), which aim to utilize the existing network of roads better and optimize traffic as well as adaptive traffic light control (ATSC), one of the main components of these systems, have become unavoidable today [4].

Traditional traffic signal control methods are not responsive to dynamic and fluctuating traffic conditions [5]. For example,

fixed-time traffic signal control (FTSC) methods regulate signal phase lengths and phase sequences from pre-programmed programs and are not capable of responding to sudden traffic flow changes, unexpected events such as accidents, or emergency vehicle requests. The wasteful cycle times utilized in these systems can result in long traffic queues and significant delays [6]. Another traditional approach, vehicle-operated traffic controllers, obtain real-time traffic data using inductive sensors and increase or decrease the green light time depending on the current scenario. Although these systems are more generalized than fixed-time control systems, they may lose effectiveness by behaving like fixed systems on high-density roads or where spacing between intersections is short [7]. On the other hand, adaptive traffic signal control (ATSC) systems; based on real-time observations of traffic parameters such as queue length, number of vehicles approaching and speed, they adapt the duration of traffic signals in real time and recalibrate the control rules continuously. ATSC systems are considered to be one of the important elements in achieving the goals of sustainable transport in intelligent cities. Of these, two are particularly salient here: fuzzy logic (FL) and reinforcement learning (RL) [8].

Fuzzy logic (FL)-based control systems have been designed to manage uncertain and dynamic road traffic flow with flexible rules similar to human thinking. They can dynamically adapt traffic signal timings based on traffic parameters, such as by defining the traffic density through linguistic variables such as "low", "medium", or "high". FL-based controllers have excellent potential in reducing the average travel time and waiting time at isolated intersections. Recent studies prove that FL systems outperform traditional methods even in situations where emergency vehicles receive priority. For example, one study proposed an FL-based intelligent traffic signal control system for an isolated four-pillar intersection; the proposed model is designed to optimize the intersection efficiency by minimizing the average waiting time of the vehicles. This system exhibited excellent improvement in the average waiting time of vehicles and the average waiting time of emergency vehicles compared to traditional fixed-time and delay control policies. Fuzzy logic is used widely in engineering, especially in traffic control, due to the fact that it possesses the ability of providing simple and implementable solutions to complex problems [1].

Reinforcement learning (RL) methods are powerful artificial intelligence tools that learn by trial and error and from

interactions with the world; they have as their general goal maximizing long-term performance. One of the biggest advantages of RL systems is that they do not require a pre-specified mathematical model for complex and dynamic systems such as road traffic. This makes RL useful in applications with dynamic and uncertain factors such as human actions, vehicle interactions, weather and unexpected collisions. In this approach, RL learns its control policy for the road system based on direct interaction with the environment and its own experience. Based on this "self-learning" or "online adaptive learning" ability, RL algorithms can update their model over time based on new observations and learn to adapt sensitively to varying road conditions [7], [9]. Three key components are found in RL-based traffic control systems: the state of the environment, available actions and the reward function. The environmental state comprises traffic conditions such as vehicle position and speed and queue length; the action comprises control commands such as signal phase selection or dynamic adjustment of its length. The reward function is typically expressed in terms of metrics such as total waiting time or negative queue length. In this case, the agent receives numeric feedback from the environment, i.e., the reward, for each of its actions. This reward helps the agent to evaluate the result of its action and to get better with the passage of time [4].

RL is becoming more common in traffic signal control due to its ability to adapt to varying and unpredictable traffic conditions, under which traditional fixed-time or vehicle-initiated control strategies are unable to properly cope with rapid traffic flow variations and unexpected incidents. Reinforcement learning-based systems attempt to maximize the capacity by minimizing performance measures such as travel time, waiting time and queue length. Among the methods of reinforcement learning, one of the most common strategies in this instance is the Q-Learning (QL) algorithm, a model-free approach. QL utilizes a trial-and-error learning method to find complex and arbitrary environments and learns its decision rule by constructing a table of "Q values" for each state-action pair. The agent learns to change these Q values at each learning step and decides on the optimal action in similar situations it encounters in subsequent steps. In a way, reinforcement learning algorithms can possibly be very powerful in solving complex decision-making problems [2], [10].

This work entails a comparative analysis of four different traffic signal control methods in an isolated intersection setting: fixed-time control, actuated control by the SUMO simulator, fuzzy logic (FL) control and reinforcement learning (RL) control. A comprehensive analysis of the influence of different training durations (length of episodes) on the performance of the RL method is included. The experiments are conducted under two scenarios of low and high traffic density simulated using two demand levels with performance measures such as average travel time and average waiting time. Results show that AI-based adaptive traffic control systems have the potential to provide more adaptive and efficient solutions for urban mobility both in dense and variable traffic situations.

The rest of this study is organized as follows: Section 2 introduces the SUMO simulation program. Section 3 discusses the traffic light control methods. Section 4 presents simulation

studies and results analysis. Finally, Section 5 presents the results of the study and information on future work.

II. SIMULATION OF URBAN MOBILITY (SUMO)

Simulation of Urban Mobility (SUMO) is a key tool for the creation of the model, simulation and testing of the performance of modern traffic management systems. By utilizing its microscopic modeling approach, SUMO is capable of simulating each vehicle individually, enabling users to learn about the characteristics of a specific traffic flow in a selected road network [11]. Using the Python API TraCI (Traffic Control Interface), dynamic control algorithms can be run interactively with SUMO, which provides a stream of real-time data. As it is open source and no cost, SUMO is widely used in academic research. It also allows flexible and quick modeling of extensive traffic networks, providing efficient modeling infrastructure, mainly for evaluating artificial intelligence-based control approaches [12].

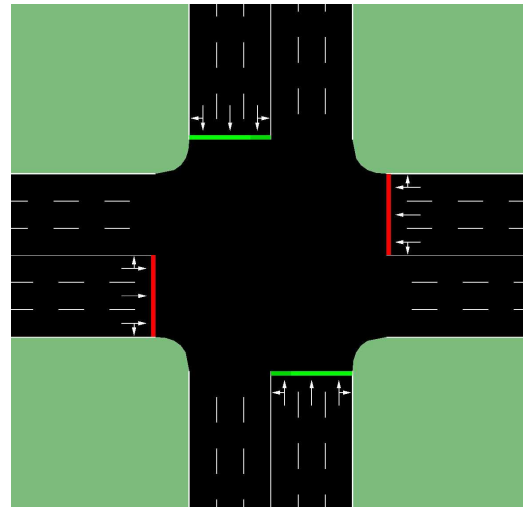


Fig. 1. SUMO Interface.

III. TRAFFIC SIGNAL CONTROL METHODS

A. Fixed-Time Signal Control

Fixed-time traffic signal control (FTSC) is among the most traditional and dominant methods of city traffic management. This approach is preprogrammed with a predetermined traffic signal program, where the duration of green, yellow and red phases for each movement is constant and does not vary according to the real traffic demand. Whereas the cycle time and phase time are usually optimized based on historical traffic knowledge or intuitive estimates, these times remain constant in the system operation [13].

Fixed-time controller has been employed as a static plan of traffic signals in this study within the SUMO simulation environment. The control process is a fixed phase time that does not vary, regardless of the existing traffic conditions or instantaneous demand for vehicles. The phase split is assigned uniformly or evenly before the simulation; no adaptive feedback mechanism is used. Despite its simplicity of implementation and computational efficiency, the fixed-time

system is inherently unresponsive to traffic variations, leading to operational inefficiencies such as increased delays, excessive congestion in queues and inefficient use of green phases, especially in congested and volatile situations.

In this context, the fixed-time control method is positioned as the baseline benchmark in this study; in comparison with this method, the performance gains and responsiveness to changing traffic demands when using more flexible and adaptive control strategies such as vehicle control, fuzzy logic and reinforcement learning are analyzed.

B. Actuated Signal Control

Actuated traffic signal control is a demand-based control strategy in which signal phase transitions are determined based on the presence of a vehicle in real time. In this approach, vehicle presence is typically detected using inductive loops or virtual sensors defined in simulation environments. Unlike fixed-time control systems, actuated systems dynamically allocate green phase durations based on the current traffic situation, allowing for more flexible and situational phase scheduling [14].

In this study, actuated control system is simulated through the intrinsic functions of the SUMO simulation environment by changing the signal phases to respond in real-time to arriving vehicles at the intersection. Under conditions of low demand on one approach, the comparable green phase duration is decreased or a phase jump is implemented, which reduces idleness and increases intersection efficiency.

Whereas actuated control is more flexible than time-fixed modes, it also relies on pre-programmed logic patterns and threshold trigger levels. This limits the system to respond less to sensor location and rule setting. This impedes the functionality to optimize signal timing as much as possible, especially for highly variable or heavy traffic conditions.

C. Fuzzy Logic Control

Fuzzy logic (FL) is an artificial intelligence that was created to deal with dynamic and uncertain traffic flow when fixed timing and actuated control are not applicable traffic signal control methods. In opposition to traditional Boolean logic, fuzzy logic adheres to the concept of "degrees of truth"; instead of just determining whether or not a condition is "true" or "false," a compromise, *i.e.*, probabilities, is also considered. This allows inference from rough values expressed as linguistic terms such as "low", "medium" and "high", facilitating modeling of realistic decision rules similar to human reasoning [1].

The Mamdani fuzzy inference method, one of the most popular and easily comprehensible fuzzy reasoning approaches, is used in this study. For traffic control applications, the Mamdani approach is particularly effective compared to other options, such as the Sugeno model. The creation of smooth output surfaces through centroid-based defuzzification and the intuitive formulation of linguistic "IF-THEN" rules are two of its main benefits, which make it perfect for simulating human-like decision-making in dynamic traffic situations [1, 3, 15]. A Mamdani-type fuzzy inference system incorporating instantaneous queue lengths in opposite directions was developed in to determine the traffic signal

green signal timing of an isolated four-lane intersection in real time. The controller implemented uses two input variables, namely the queue length in the current direction and the queue length in the opposite direction. Both variables are limited to the range [0, 50] for a car and are characterized by three triangular membership functions with adjectives "low", "medium" and "high". The output variable, the green light duration, is between 5 and 50 seconds and is characterized by five linguistic terms: "very short", "short", "medium", "long" and "very long". These terms are also represented by triangular membership functions. The membership functions for the inputs and outputs of the fuzzy logic control system are shown in Figures 2 and 3, respectively.

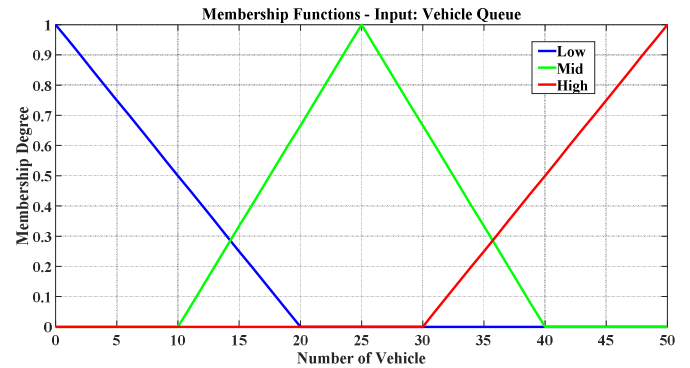


Fig. 2. Input Membership Functions.

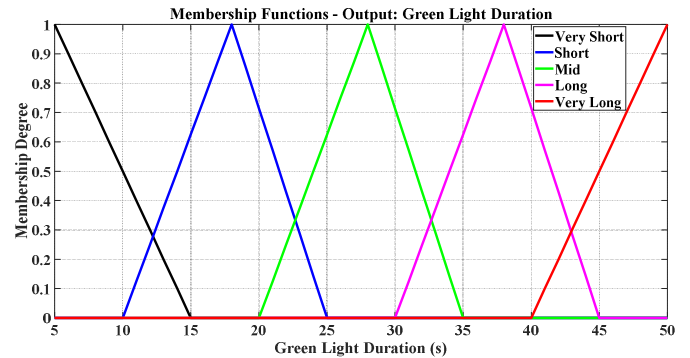


Fig. 3. Output Membership Functions.

A total of nine fuzzy control rules is defined, reflecting the combinations of traffic density between two conflicting directions. For example, when high traffic density is detected in the current direction and low traffic density in the opposite direction, the controller assigns a "too long" green signal time to the corresponding direction. The defined rules are listed in Table 1. The fuzzy logic controller operates independently for both directions in each signal update cycle. First, an appropriate green signal time is calculated for the north-south direction, then the same process is repeated for the east-west direction by changing the input queues. The final phase plan, including the yellow signal transition time, is dynamically applied to the SUMO simulation environment via the TraCI API.

TABLE I
FUZZY LOGIC RULES

Rule Number	Related Queue	Other Queue	Green Light Duration
1	High	Low	Very Long
2	High	Mid	Long
3	High	High	Mid
4	Mid	Low	Mid
5	Mid	Mid	Short
6	Mid	High	Very Short
7	Low	Low	Short
8	Low	Mid	Very Short
9	Low	High	Very Short

This approach enables real-time adaptation without relying on pre-set thresholds or rigid time schedules and provides increased flexibility in managing asymmetric and unstable traffic flows, especially at isolated intersections.

D. Reinforcement Learning Based Control

Reinforcement learning (RL) offers a powerful framework for solving sequential decision-making problems in dynamic, uncertain and partially observable environments. Such structures are widespread in urban transportation networks. In traffic light control, RL involves a learning process in which an agent continuously interacts with the road system, observes the current situation and selects actions that attempt to minimize congestion-related performance metrics such as vehicle delay, queue length and waiting time. Unlike rule-based or time-fixed systems, RL does not utilize pre-defined heuristics or invariant rules, but instead its policy evolves over time by learning from reward-based feedback in the environment. Due to this data-driven adaptive model, RL stands out with its capability to manage real-time control demands efficiently, especially in heavy or unstable traffic situations.

In this study, an RL-based traffic signal controller for a single intersection is implemented. The environment is simulated by the SUMO traffic simulator, with which the agent constantly interacts in real time through the TraCI (Traffic Control Interface). The primary function of the controller is to dynamically manipulate the green phase length of the signal for the two conflicting directions: north-south (NS) and east-west (EW) depending on the encountered traffic conditions.

RL controller was utilized using the standard Q-learning algorithm, which is based on the time-difference rule and doesn't require a model to estimate the optimal action-value function. Individual Q-learning agents were trained in each travel direction (north-south [NS] and east-west [EW]). Each agent chooses from a finite set of actions represented by possible green phase durations: {10, 20, 30, 40} seconds. The state space of the system is defined in terms of levels of queue length using real-time traffic data. There are three linguistic states (low, medium and high) per direction of the queue length, resulting in a total of nine combinations of states. The topology allows adaptive decision-making by agents using the state of the traffic. To balance exploration and exploitation in strategy selection, a greedy strategy with $\epsilon = 0.1$ was implemented. The learning rate for updating the Q-value was set to $\alpha = 0.1$ and the discount rate used to estimate future

rewards was set to $\gamma = 0.9$. These parameters allow the system to learn quickly and consider long-term optimization.

The reward function defined for the RL-based controller is tuned to reduce the total waiting time of a vehicle in the system. The instantaneous reward calculated at each decision stage is defined as the difference between the total waiting time at the previous and current stages on all approaches (lanes). This structure incentivizes agents to perform actions aimed at reducing traffic congestion.

The green phase duration chosen by the Q-learning agents was dynamically applied to the intersection via the Traffic Control Interface (TraCI) at each traffic light cycle. The simulation was run for a full traffic light cycle in each episode, maintaining constant interaction with the environment. The training process was carried out for 500 episodes; updated Q-values were recorded at the end of each episode and the resulting Q-table was exported for performance evaluation under different traffic scenarios.

IV. SIMULATION AND RESULTS

The simulation environment was structured by modeling an isolated signalized intersection with four branches, each representing one of the main traffic directions. All experiments, conducted using the SUMO traffic simulator, were conducted in a common network topology, ensuring experimental consistency when comparing control methods.

This study considered two separate traffic scenarios to evaluate the effectiveness of traffic control strategies under different densities. The low-density scenario involved approximately 3,200 vehicles, while the high-density scenario involved approximately 6,700 vehicles. Vehicle departure times were randomly determined for a fixed simulation period and flows were determined according to probability distributions. All vehicles were classified as passenger cars and were allowed to enter the intersection from any available lane. Traffic flow at the intersection was modeled as bidirectional and signal phasing decisions were made independently for the north-south (NS) and east-west (EW) directions.

To ensure the integrity of the experiment, the signal phase patterns, yellow signal switching times, sensor placement and vehicle routing strategies were kept constant for all control methods. This consistent configuration allows direct and meaningful comparison of fixed-time, actuated, fuzzy logic and reinforcement learning control strategies under different traffic conditions.

A. Low Traffic Density

The results of the first performance metric, average waiting time, under low traffic density conditions are presented in Figure 4. Analysis of the results reveals significant differences between the control strategies. The most successful method in terms of average vehicle waiting time was the actuated control method with a value of 3.40 seconds, which is 76.8% better than the fixed-time method. The fuzzy logic controller achieved a 42.4% improvement with a value of 8.44 seconds, while RL model trained on 300 episodes achieved similar success, achieving a reduction of 44.0%. In contrast, the other RL variants performed worse. The model with 150 episodes showed an improvement of only 3.6%, while the models with

50 and 500 episodes showed higher waiting times. This suggests that RL algorithms require careful tuning to achieve adequate generalization in low traffic density conditions.

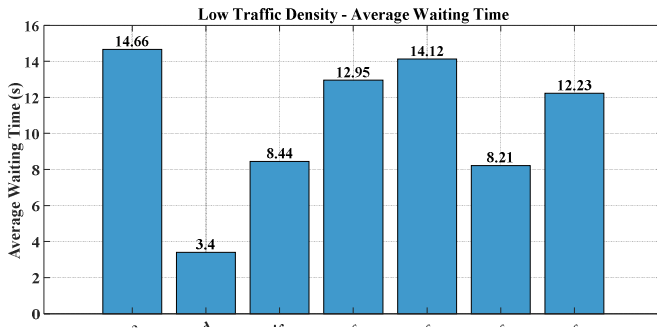


Fig. 4. Average Waiting Time for Low Traffic Scenario.

According to the graph in Figure 5, the best performance in terms of average travel time under low congestion conditions was achieved by the actuated control method, reaching 26.17 seconds; this value represents a 31.4% improvement over the fixed-time method. The fuzzy logic and RL models trained on 300 episodes also achieved about a 17% reduction in time, with results of 31.68 and 31.56 seconds, respectively. In contrast, the RL controller trained on 150 episodes lagged behind even the fixed-time method, with a result of 38.25 seconds. The other RL variants, however, showed mediocre results, failing to compete with simpler systems. These results indicate that responsiveness is critical for time optimization under low congestion conditions.

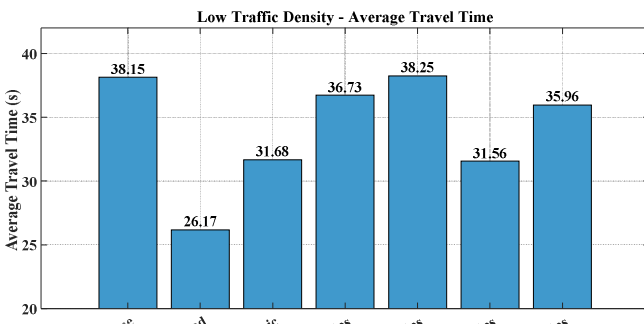


Fig. 5. Average Travel Time for Low Traffic Scenario.

B. High Traffic Density

Figure 6 presents the results of the average waiting time obtained at the conclusion of the study conducted under high-traffic conditions. The best result of 44.20 seconds was achieved by RL controller trained on 300 episodes. This result is 28.3% better than the fixed-time controller. Moreover, other RL models trained on 50 and 500 episodes also demonstrated competitive results with values of 47.80 and 49.18 seconds, respectively. In contrast, the fuzzy logic controller with a value of 55.52 seconds provided some improvement over the fixed-time method but was inferior to the RL systems. The fixed-time and actuated controllers with results of 61.60 and 61.43 seconds, respectively, demonstrated poor control performance under heavy traffic conditions, indicating limited adaptability. The results show that RL based control systems supported by

appropriate training are more effective in heavy traffic conditions and can significantly outperform traditional methods in terms of reducing waiting time.

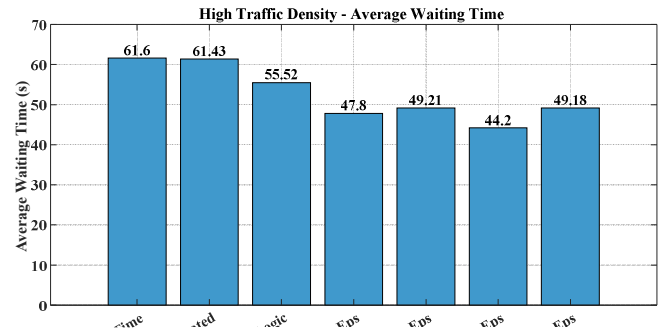


Fig. 6. Average Waiting Time for High Traffic Scenario.

According to the graph in Figure 7, in the high-traffic scenario, RL controller trained on 300 episodes performed the best, with an average travel time of 76.06 seconds. This value represents a 15.2% improvement over the fixed-time controller, demonstrating that RL algorithm can effectively adapt to high-traffic conditions. In addition, RL models trained on 50 and 150 episodes performed better than the classical methods, with values of 79.39 and 81.55 seconds, respectively. Although not as optimal as the 300-episodes model, they demonstrate the advantage of learning-based adaptation over static control logic. On the other hand, the fuzzy logic controller, with an average time of 86.72 seconds, performed slightly better than the fixed-time controller (89.71 seconds) and the actuated controller (89 seconds) but was outperformed by RL approaches. These results indicate that RL systems, especially those supported by longer training periods, can provide significant improvements in overall driving efficiency by more accurately adapting signal activation times in high traffic conditions.

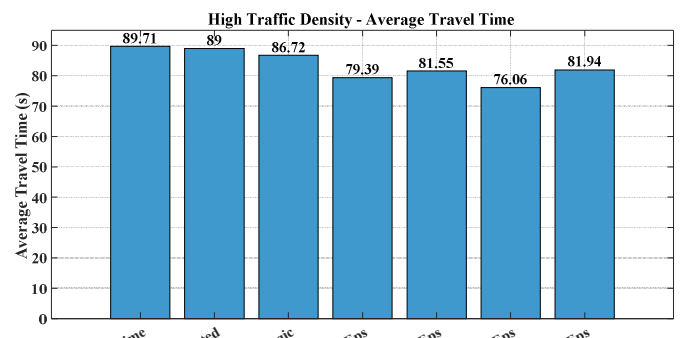


Fig. 7. Average Travel Time for High Traffic Scenario.

A comparative analysis of both traffic situations revealed apparent differences in performance among the utilized traffic signal control strategies. Under low-density traffic situations, the actuated control strategy performed best with shorter waiting and moving times made possible through real-time response. Fuzzy logic and 300-training-episode RL controllers also did well; however, RL was extremely sensitive to the number of segments under this configuration and some configurations lagged significantly behind the static methods.

In conditions of high density, RL-based control systems performed best out of all techniques, especially when training time was sufficient. The 300-episode model worked best in moving and waiting times, reflecting the benefit of learning frameworks in dense and complex traffic flows. In this sense, fixed-time and RL control schemes performed poorly due to their failure to respond to increasing traffic volume.

Overall, the results show that the most effective traffic signal control method is highly dependent on traffic density. While reactive and simple systems may suffice in low-density situations, adaptive controllers with learning techniques, such as RL, offer better solutions, especially under more challenging and dense traffic conditions.

V. CONCLUSIONS

In this study, four control strategies for traffic lights—fixed time, fuzzy logic, actuated, and reinforcement learning—were compared in terms of performance using the SUMO traffic simulator at a single intersection. Performance was measured in terms of average waiting time and average travel time and adaptability and efficiency of the strategies at different levels of traffic density were compared.

The simulation results showed that the performance of the control strategies significantly depends on the traffic density. Under low-density conditions, the actuated control method yielded the most effective results due to its structure, which directly responds to the presence of a vehicle. On the other hand, fuzzy logic-based controllers and moderately trained reinforcement learning models yielded promising results in some scenarios, but exhibited fluctuations in stability. In contrast, reinforcement learning-based strategies demonstrated clear superiority in high-density traffic conditions, especially after completing enough training cycles, e.g. 300 segments, achieving the lowest values for both performance metrics. This clearly demonstrates the potential of RL-based controllers for efficient congestion adaptation and dynamic phase timing optimization.

In this context, the study represents a pioneering assessment of the potential contribution of RL-based control strategies not only in the case of individual intersections, but also when moving to multi-intersection systems. Further research aims to develop AI-based traffic management systems that optimize network-wide performance by facilitating coordination and information exchange among controllers within multi-agent system architectures. Specifically, upcoming studies will expand the suggested framework to more complex and interconnected traffic networks to assess the scalability and adaptability of the control strategies under actual urban conditions. Additionally, experimental validation will be sought to supplement simulation-based results, and the incorporation of statistical analysis techniques will be considered to enhance the robustness of performance evaluation.

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